Gymnasium Taxi

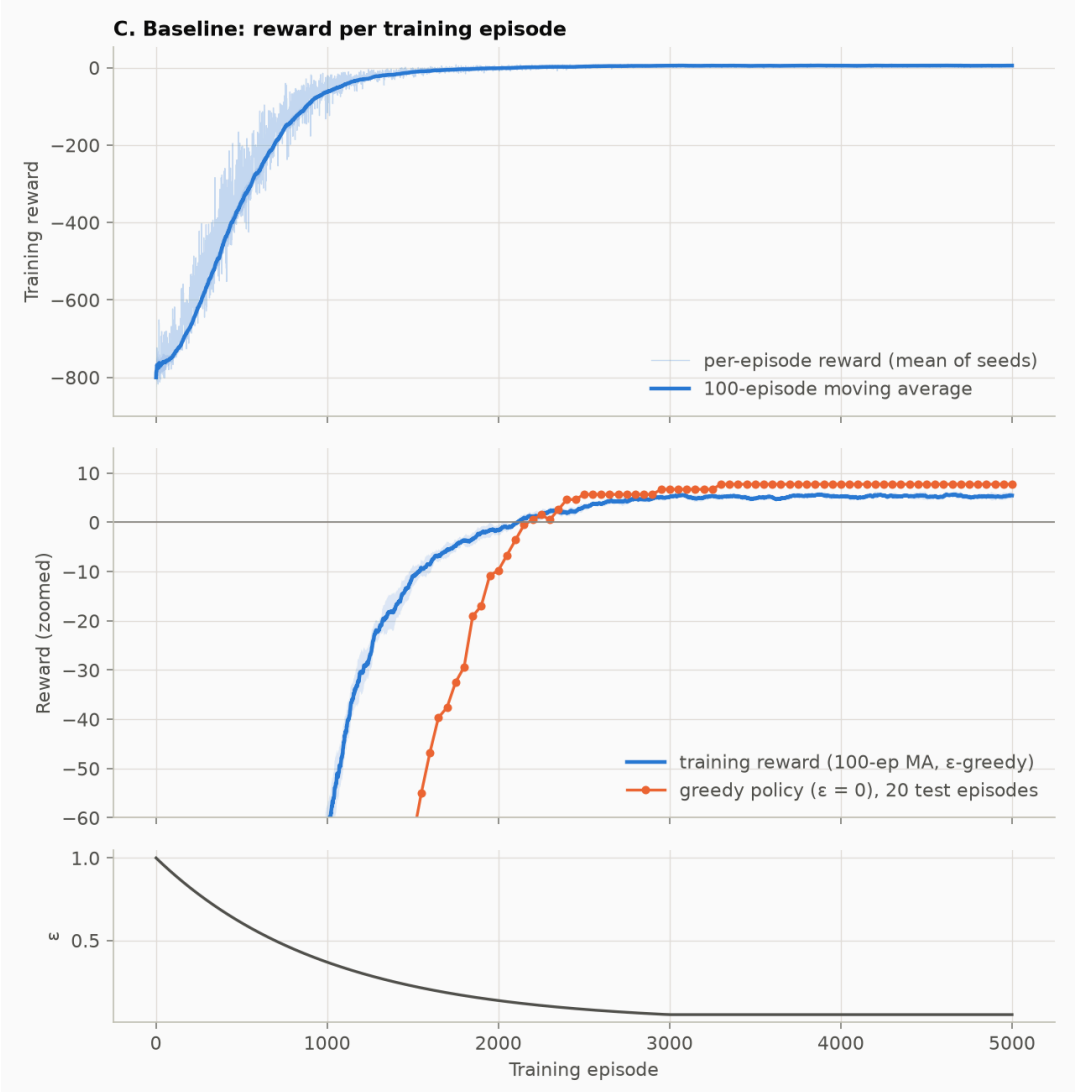
Environment-

The environment is a taxi on a 5x5 grid with walls that must drive to the passenger waiting at a few different spots. The driver picks them up then drops them off at their destination. When it comes to the states it's 25 taxi cells times 5 different locations including in taxi times 4 destinations. There's 6 actions and the transitions are deterministic. The program terminates on delivery.

Q-learning implementation-

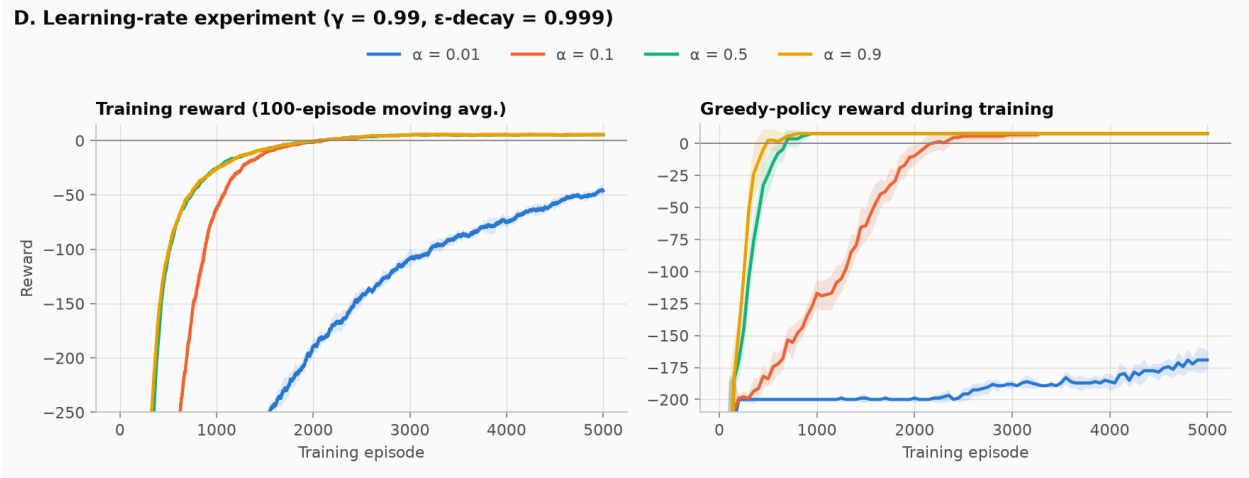
When it comes to the terminal handling there's no future value so the target is just r. A 200 step timeout isn't a real terminal state so it bootstraps from s. Each setting is trained for 10 seconds. The greedy policy E = 0 is scored every 50 episodes

Baseline-



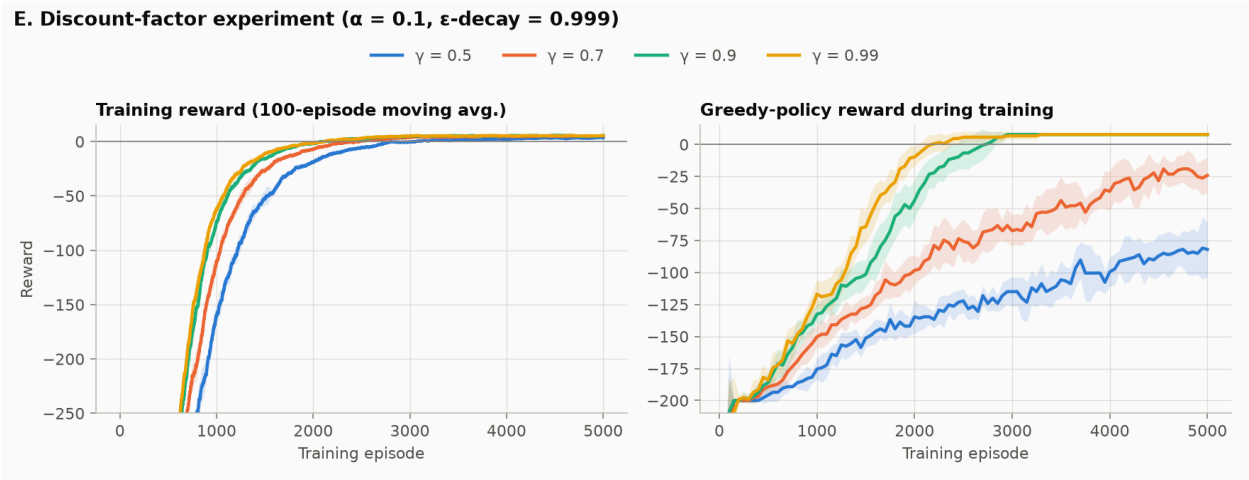
The 100 episode average crosses 0 near episode 2050 and it becomes near optimal around 2,370.

Learning rate-

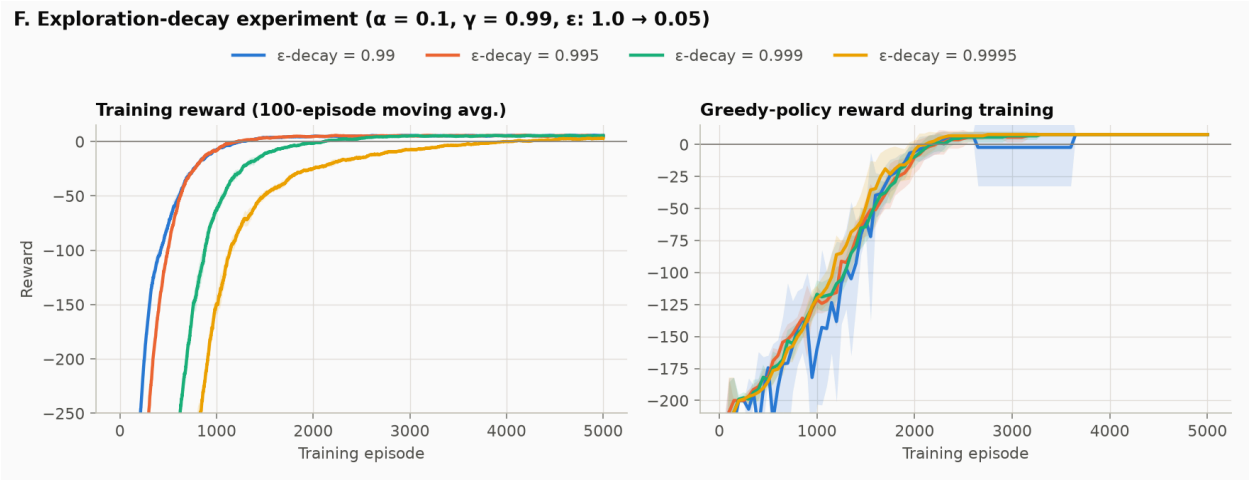


Taxi is deterministic

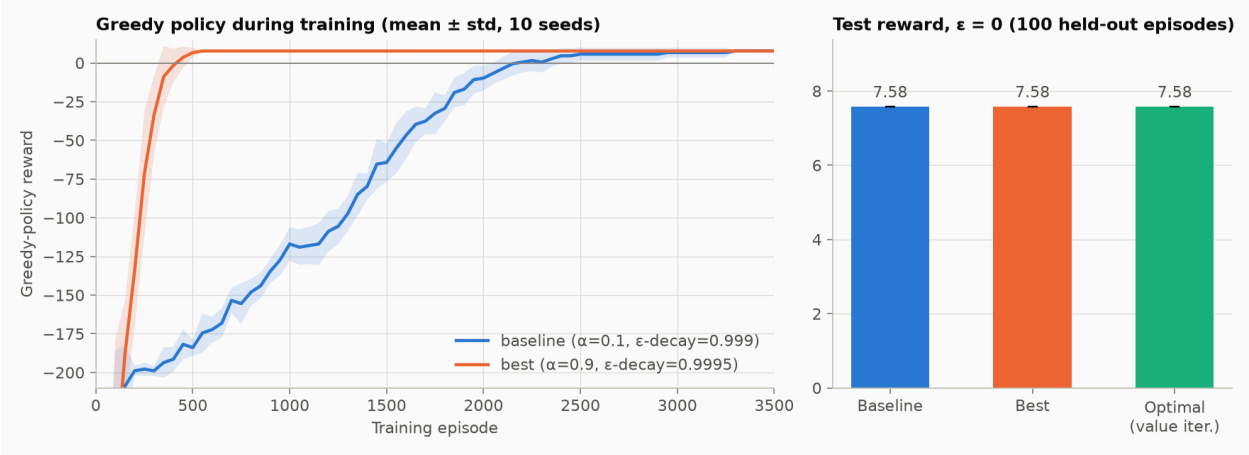
Discount factor-



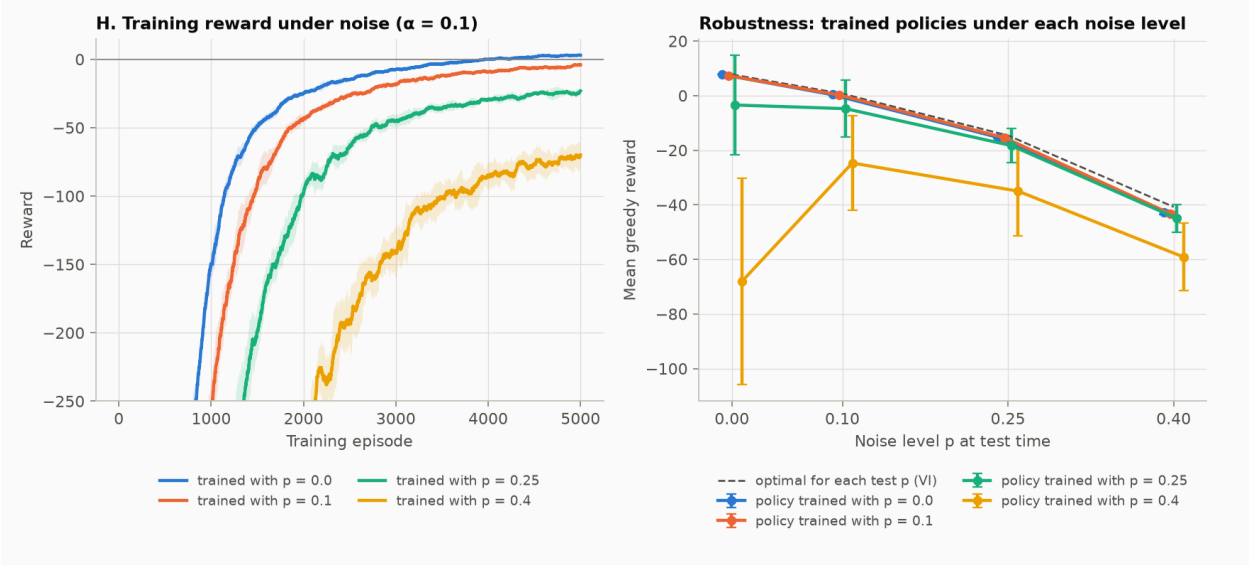
Decay-



Best agent-



Stochastic environment-



Conclusions-

- Q-learning solves taxi as well as the tuned agent matches the optimal policy on the test episodes
- a controls speed and high a is fastest when deterministic but has to be small under noise
- Y has to be close to 1 so that the delivery reaches the start of the step route
- E decay trades training reward for an accurate q table